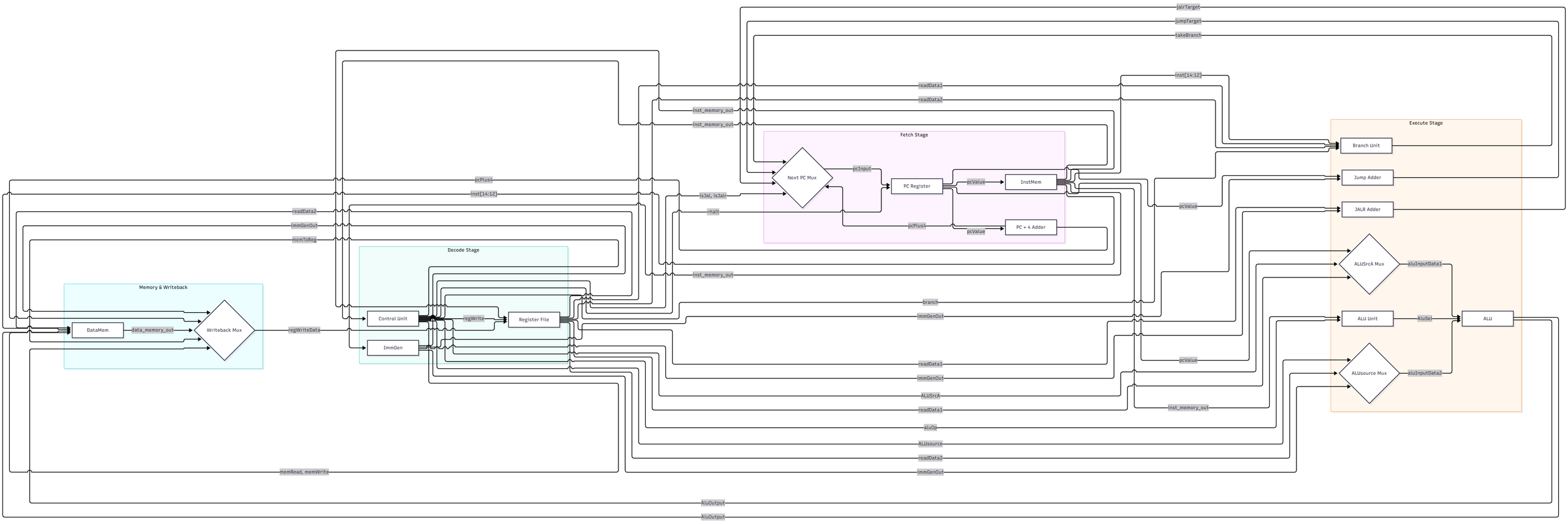

jalrTarget
jumpTarget
takeBranch
inst[14:12]
readData1
readData2
inst_memory_out
inst_memory_out
Fetch Stage
Next PC Mux
pcInput
PC Register
pcValue
InstMem
pcPlus4
PC + 4 Adder
pcValue
Execute Stage
Branch Unit
Jump Adder
JALR Adder
pcPlus4
inst[14:12]
isJal, isJalr
halt
pcValue
readData2
immGenOut
memToReg
Decode Stage
ALUSrcA Mux
aluInputData1
inst_memory_out
Memory & Writeback
Control Unit
regWrite
Register File
branch
immGenOut
DataMem
data_memory_out
Writeback Mux
regWriteData
ImmGen
ALU Unit
AluSel
ALU
pcValue
readData1
immGenOut
ALUSrcA
ALUsource Mux
aluInputData2
inst_memory_out
readData1
aluOp
ALUsource
readData2
immGenOut
memRead, memWrite
AluOutput
AluOutput